

JOSH MCCAMLEY

TECHNICAL GAME DESIGNER

PORTFOLIO

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EDUCATION

BACHELOR OF ARTS (HONS) GAMES DESIGN

Ulster University, Belfast, N-Ireland

- ❖ First Class Honours; Achieved **Highest Grade** in the graduating cohort.
- ❖ **Enterprise Placement Year (EPY)**, securing a **Distinction** and Diploma in Professional Practice.

KEY SKILLS

SOFTWARE

- ❖ Unreal Engine 5
- ❖ C++
- ❖ Visual Scripting (Blueprints)

GAMEPLAY & SYSTEMS

- ❖ Systems Architecture
- ❖ AI (Behavior Trees, EQS, A*)
- ❖ Data-Driven Design

PIPELINE & INFRASTRUCTURE

- ❖ Tool Creation & Editor Utilities
- ❖ Perforce / Git Version Control
- ❖ Optimisation & Profiling

DESIGN & WORKFLOW

- ❖ Rapid Prototyping
- ❖ Systems Troubleshooting
- ❖ Technical Documentation
- ❖ Flow Diagramming

ABOUT ME

Technical Game Designer with 3+ years of experience in bridging high level design with C++ and Visual Scripting. My bread and butter is building responsive game loops and scalable, data driven systems. Behind the scenes, I am obsessed with removing pipeline bottlenecks - taking immense pride in building custom tools and accessible subsystems that empower teams to iterate without friction, validated by successfully securing £60k in studio funding.

EXPERIENCE

TECHNICAL GAME DESIGNER | JUNE 2023 - PRESENT | MAD MOON STUDIOS

Turned an Academic Placement year into a funded studio, temporarily leaving University to demo projects all over the world.

- ❖ Secured **£60,000 incubation funding** from Northern Ireland Screen through successful pitching and comprehensive **technical documentation**; managed a team of 3 and **collaborated** with an external producer to **consistently hit development milestones**.
- ❖ Implemented a **FLECS based** projectile system in C++, achieving a **131x improvement** in CPU Game Thread time (simulating 100,000 projectiles) while exposing the logic to **Visual Scripting for team-wide usability**.
- ❖ Designed and Scripted a flying enemy ecosystem using **A* pathfinding** and **EQS**, enabling dynamic evasion maneuvers **responsive to environmental geometry and combat thresholds**, all managed by **Behaviour Trees**.
- ❖ Programmed a persistent **Quest Subsystem** decoupled from the engine via auto-reimporting CSVs, empowering the team to **rapidly edit** questlines, objectives, and rewards **externally without relying on an engine instance**.
- ❖ Designed a **runtime editable, grid based inventory** system, paired with in-engine tooling to instantly load items by gameplay tag into a property matrix, **drastically accelerating crafting implementation and economy balancing**.
- ❖ Developed a fully **modular, replicated** player character featuring a 6DOF movement controller, utilising Anim Blueprints and procedural animations to **ease character asset iteration**.

PROJECTS

PROCTEX - AN UNREAL ENGINE PLUGIN

- ❖ Built a custom UE5 plugin with native UI integration for **real time 3D previews** and **procedural texture baking**, eliminating iteration **bottlenecks** in the studio's technical art pipeline.

SELF-HOSTED DEVELOPMENT INFRASTRUCTURE

- ❖ Deployed a centralised Nextcloud and Perforce server environment **tailored to the team's pipeline**, easing asset management and establishing a **reliable**, autonomous version control.

PERFORCE LIVE FEED APP

- ❖ Developed a custom Nextcloud app that provides a **fully searchable, live feed** of checked out files and pending changelists, **minimising** workflow overlap and accelerating file debugging and diffing, instantly becoming a **daily tool the team relies upon**.